

Chapter 6: 4th and 5th editions
Section 6.1, pp. 161-163, Nos. 5, 6, 7

Section 6.2, p. 169, Nos. 4

5) Eliminate all useless productions from the grammar

$S \rightarrow aS|AB,$
 $A \rightarrow bA,$
 $B \rightarrow AA.$

Step 1: Remove Nullable Variables

$S \rightarrow aS|A \rightarrow S \rightarrow aS \rightarrow S$

Step 2: Remove Unit-Productions

$S \rightarrow \phi$

Step 3: Remove Useless Variables

ϕ

What language does this grammar generate?

Since none of the variables in the grammar generate a string then the language generated by this grammar has no string. Hence, this grammar generates an empty language ϕ ,
 $L(G) = \phi$;

6) Eliminate useless productions from

$S \rightarrow a|aA|B|C,$
 $A \rightarrow aB|\lambda,$
 $B \rightarrow Aa,$
 $C \rightarrow cCD,$
 $D \rightarrow ddd,$

Step 1: Remove Nullable Variables

$S \rightarrow a|aA|B;$
 $A \rightarrow aB|\lambda;$
 $B \rightarrow Aa;$
 $D \rightarrow ddd$

Step 2: Remove Unit-Productions.

$S \rightarrow a|aA|Aa;$
 $A \rightarrow aB|\lambda;$
 $D \rightarrow ddd;$

Step 3: Remove the unreachable variables.

$S \rightarrow a|aA|Aa;$
 $A \rightarrow aB|\lambda;$

7) Eliminate all useless productions from

S $\rightarrow$ **AaB|aaB**
A $\rightarrow$ λ ,
B $\rightarrow$ **bbA| λ .**

Step 1: Remove Nullable Variables

S $\rightarrow$ **AaB|aaB**
A $\rightarrow$ λ ,
B $\rightarrow$ **bb| λ .**

Step 2: Remove Unit-Productions.

S $\rightarrow$ **AaB|aaB**
A $\rightarrow$ λ ,
B $\rightarrow$ **bb| λ .**

Step 3: Remove the unreachable variables.

Note: A is reachable but unnecessary since is not affecting the grammar in any way.

S $\rightarrow$ **aB|aaB**
B $\rightarrow$ **bb| λ .**

4) Transform the grammar with productions

S $\rightarrow$ **abAB,**
A $\rightarrow$ **bAB| λ ,**
B $\rightarrow$ **Baa|A| λ**

First: Removing λ productions:

S $\rightarrow$ **abAB|abA|abB|ab**
A $\rightarrow$ **bAB|bA|bB|b**
B $\rightarrow$ **Baa| bAB|bA|bB|b|aa**

Second: Introduce variables C and D to substitute terminals a and b.

S $\rightarrow$ **CDAB|CDA|CDB|ab**
A $\rightarrow$ **CAB|bA|bB|b**
B $\rightarrow$ **BCC| bAB|bA|bB|b|CC**
C $\rightarrow$ **a**
D $\rightarrow$ **b**

Third: Introduce variables to shorten the right sides of the production.

S $\rightarrow$ **EF|EA|EB|CD**
A $\rightarrow$ **CF|bA|bB|b**
B $\rightarrow$ **BG|bF|bA|bB|b|CC**
E $\rightarrow$ **CD**
F $\rightarrow$ **AB**
G $\rightarrow$ **CC**
C $\rightarrow$ **a**
D $\rightarrow$ **b**